

CISCO VIRTUAL INTERNSHIP



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INSTITUTION - IILM UNIVERSITY GREATER NOIDA

PROGRAM - B.TECH CSE

SEMESTER - 5TH

ORGANIZATION – CISCO NETWORKING ACADEMY

Internship & Course Overview

- The internship focused on basic networking and cybersecurity concepts.
- Three courses were completed through Cisco Networking Academy.
- Packet Tracer was used to understand and practice network simulation.
- Cybersecurity Essentials introduced threats, protection, risk, and response.
- Together, the courses provide a foundation for further networking and cybersecurity study.

LEARNING PATH

1 01 Getting Started

Learn the Packet Tracer environment.

2 02 Exploring Networking

Build and observe a small network.

3 03 Cybersecurity Essentials

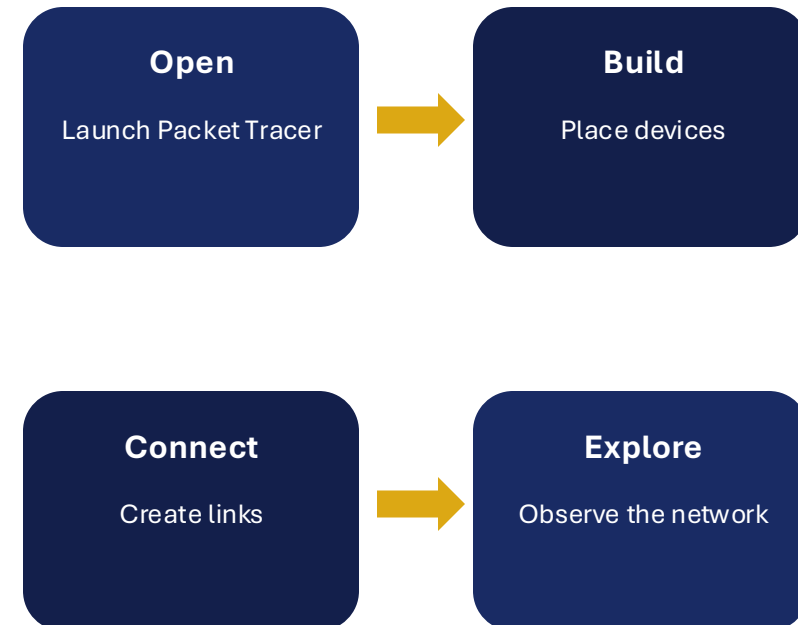
Understand threats, defense, and response.

Getting Started with Cisco Packet Tracer

COURSE PURPOSE

- Introduces Cisco Packet Tracer as a network simulation and visualization tool.
- Explains the basic interface, workspace, and device selection.
- Shows how to create a simple network without physical hardware.
- Introduces basic connections and interaction with network devices.
- Provides a starting point for practical networking activities.

FROM TOOL → TO PRACTICE



Exploring Networking with Cisco Packet Tracer

PRACTICAL NETWORKING

- Builds and explores a small office network using Packet Tracer.
- Connects devices such as PCs, switches, routers, and wireless components.
- Uses simulation to observe how packets move through the network.
- Introduces basic network management and configuration concepts.
- Encourages troubleshooting by checking connections, configuration, and communication.

SIMPLE NETWORK VIEW



Simulation → packet flow → configuration → troubleshooting

Cybersecurity Essentials — Core Concepts

MAIN AREAS

E

Endpoint Security

Protect computers, operating systems, and user devices.

N

Network Defense

Protect network communication and control access.

T

Threat Management

Identify threats, assess risk, and support response.

- Security is based on protecting confidentiality, integrity, and availability.
- The course develops a basic understanding of how attacks happen and how defenses reduce risk.

Cybersecurity Essentials – Threats & Attacks

COMMON THREATS

M

Malware

Malicious software that can damage, disrupt, or gain unauthorized access.

P

Phishing

Deceptive messages used to steal information or deliver malicious content.

S

Social Engineering

Manipulating people into revealing information or taking unsafe actions.

D

DoS / DDoS

Overwhelming a service so legitimate users cannot access it normally.

W

Weaknesses

Weak configurations, software flaws, and poor security practices can be exploited.

Cybersecurity Essentials — Defense & Protection

KEY SECURITY CONCEPTS

1

Access Control

Only authorized users and devices should access resources.

2

Firewalls

Control network traffic using security rules.

3

Cryptography

Protect information using encryption and hashing.

4

Vulnerability Assessment

Identify weaknesses that could be exploited.

5

Risk Management

Understand threats and impact, then choose suitable controls.

6

Incident Response

Analyze suspicious activity and contain or recover from incidents.

Networking Skills Practiced

1

Network Devices

Understand the role of PCs, switches, routers, and wireless components.

2

Topologies

Create logical network layouts in Packet Tracer.

3

Communication

Understand basic device-to-device communication.

4

Packet Flow

Observe traffic movement using simulation.

5

Configuration

Practice basic network configuration and connectivity checks.

6

Troubleshooting

Use connections, configuration, and tests to locate simple faults.

Skills Gained & Connection to IDS

LEARNING OUTCOMES

- Built a stronger foundation in computer networking.
- Developed practical familiarity with network simulation.
- Learned common cybersecurity threats and basic defense mechanisms.
- Understood the importance of monitoring, risk assessment, and incident response.

CONNECTION TO IDS

1

1. Normal Traffic

Networking knowledge helps explain how devices normally communicate.

2

2. Suspicious Activity

Cybersecurity knowledge helps recognize behavior that may indicate a threat.

3

3. IDS Alerts

Together, these concepts support understanding how an IDS observes traffic and detects suspicious behavior.

Conclusion & Learning Outcomes

KEY TAKEAWAYS

- The three courses provided a connected introduction to networking and cybersecurity.
- Packet Tracer gave practical experience with network simulation and basic troubleshooting.
- Cybersecurity Essentials introduced threats, vulnerabilities, protection, and response.
- The knowledge gained can support further learning in network security, SOC operations, and intrusion detection.
- The courses form a useful foundation for more advanced networking and cybersecurity certifications.

OVERALL

LEARNING OUTCOME

A practical starting foundation for understanding, building, and securing computer networks.